

Julie Biedermann

☎ +33 6 47 65 19 10 | ✉ julie.biedermann@astro.unistra.fr

🌐 Julie Biedermann | 🎧 Julie-Bdm | 🆔 0009-0000-5831-3518

📍 Observatoire Astronomique de Strasbourg, Strasbourg, France - 67000

RESEARCH INTERESTS

I work on quasar and active galactic nucleus astrophysics, with a focus on observational spectropolarimetry and polarization. I have particular interests in the structure and geometry of quasars, extreme quasar systems, and candidates for supermassive black hole binaries, as well as the potential of future space-based facilities to probe quasar physics.

EXPERIENCE

■ PhD projects

October 2024 - September 2027 (expected)

Observatoire Astronomique de Strasbourg

Strasbourg, France

- Advisor : Frédéric Marin
- Study of quasars polarization in the context of the Habitable Worlds Observatory area.

Research on active galactic nuclei using optical spectropolarization, combining archival data analysis and observation obtain with ground bases facilities. Development of Python based tools for spectropolarimetric data analysis, spectral decomposition, model fitting, statistical analysis, and uncertainty estimation. Preparation and submission of ESO and SALT observing proposals. Principal Investigator of 1 observing proposal accepted by SALT.

■ Undergraduate research project

February 2024 - June 2024

Observatoire Astronomique de Strasbourg

Strasbourg, France

- Advisor : Frédéric Marin
- Is there a supermassive black hole binary system in Markarian 231 ?

Investigation of a potential supermassive black hole binary in a quasar using archival multi wavelength spectropolarimetric data. Development of a multi component spectral energy distribution model. Fitting the model to the observed SED and investigate whether the resulting model could reproduce the observed variations in polarization.

■ Undergraduate research project

May 2023 - June 2023

INS2P3 - Institut national de physique nucléaire et de physique des particules

Villeurbanne, France

- Advisor: Florian Ruppin
- Impact of the astrophysical environment of galaxy clusters on the properties of type Ia Supernovae

Study of the environmental dependence of Type Ia supernovae properties using the Zwicky Transient Facility (ZTF) sample and a galaxy cluster catalog compiled from the MCXC, Planck, and Wen catalogs. Statistical analysis and MCMC modeling of the stretch distributions of SNe Ia in cluster and in the field, including python based catalog cross matching and investigation of the resulting cosmological implications.

EDUCATION

■ PhD of astrophysics

2024 - present

Observatoire Astronomique de Strasbourg

Strasbourg, France

■ Master in Fundamental and Applied Physics

2022 - 2024

University Lyon 1 Claude Bernard

Villeurbanne, France

Cosmology and the high energy universe program with quantum field theory option.

SEMINAR AND CONFERENCES

HWO - Visionary Science and Transformational Technology Talk	December 2026 Paris, France
CNES - JC2 Poster and short presentation	October 2026 Toulouse, France
Supermassive Black Holes and Blue Notes Talk	June 2026 Montréal, Canada
YAGN - Young Astronomers on Galactic Nuclei Talk	September 2025 Southampton, UK
HWO - Visionary Science and Transformational Technology Virtual poster	July 2025 Washington DC, USA
EAS 2025 Poster and short presentation	June 2025 Cork, Ireland
SF2A 2025 - Talk Talk	June 2025 Toulouse, France
POLLUX@HWO - Talk Talk	January 2025 Paris, France

OBSERVATIONAL PROPOSALS

As Principal Investigator

7h of observing time with RSS/SALT - Accepted	2026
<i>Exploring polarimetric signature of the binary supermassive black candidate PG 1302-102 with SALT/RSS spectropolarimetry</i>	

TEACHING

■ Mathematics tutorials	24h - January 2026 - May 2026
Undergraduate students at the University of Strasbourg	

OUTREACH

■ Sciences, un métier de femmes	6 mars 2026
Outreach for high school students from the Bas-Rhin region, presenting my PhD research and academic career path and discussing opportunities and challenges for women in science for the day "Sciences, un métier de femmes" organized by <i>Femmes & Sciences</i> .	